NPTEL Online Certification Course
Industrial Engineering and Operations Research
Assignment Number 3: Detailed Solution
Indian Institute of Technology Guwahati

Q.1. Which of the following are the axioms of Total Quality Management (TQM)?

(A) Commitment (B) Scientific knowledge (C) Involvement (D) Short-term profit maximization

Sol. The axioms of Total Quality Management (TQM) are as follows: a commitment to continuous quality improvement and innovation. Scientific knowledge of the proper tools and techniques for the 'technical change'. Involvement of all in one team for the social change. TQM is long-term-oriented, not short-term, profit-driven. Hence, option (A), (B) and (C) are correct.

Q.2. Who is known as the father of quality management?

(A) W.E. Deming (B) J.M. Juran (C) P.B. Crosby (D) A.V. Feigenbaum

Sol. W. Edwards Deming is widely regarded as the Father of Quality Management. He introduced key ideas like statistical quality control, continuous improvement (PDCA cycle), and system thinking in organisations. Hence, option (A) is correct.

Q.3. Concept of zero defect was provided by

(A) W.E. Deming (B) J.M. Juran (C) P.B. Crosby (D) A.V. Feigenbaum

Sol. The concept of zero Defect was introduced by Philip B. Crosby as part of his quality philosophy. He believed that quality should be built into processes, and companies should aim for zero defects. Hence option (C) is correct.

Q.4. Who among the following has won the Deming Prize twice?

(A) W.E. Deming (B) J.M. Juran (C) P.B. Crosby (D) G. Taguchi

Sol. Genichi Taguchi won the Deming Prize twice. He first won it in 1987 for his contributions to quality control and his development of robust design methodology. He won it again in 1991 for his continued work and advancements in quality management. The Deming Prize is a prestigious award given for contributions to quality management, and Taguchi's work significantly influenced the field. Hence, option (D) is correct.

Q.5. Which of the following statement(s) is/are TRUE?

(A) A parameter is a value that describes a population.
(B) Statistics is a value that describes a sample.

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(C) A goal of inferential statistics is to generalize from samples to populations.
(D) A representative sample is a sample whose characteristics are dissimilar to the population.

Sol. A parameter is a numerical measure that describes an entire population. A statistic is a numerical measure that describes a sample taken from a population. Inferential statistics involves drawing conclusions or making predictions about a population based on information obtained from a sample. A representative sample should closely resemble or reflect the characteristics of the population, not be dissimilar. Hence, options (A), (B) and (C) are correct.

Q.6. The true average height of plants in a nursery is 25.5 cm. A random sample of 5 plants has the following heights (in cm): 24.8, 26.2, 25.7, 24.9, 26.4. The magnitude of random error in estimating the mean height is

(A) 0.5 cm (B) 0.4 cm (C) 0.3 cm (D) 0.1 cm

Sol.

The sample mean can be estimated as $\bar{x} = \frac{24.8 + 26.2 + 25.7 + 24.9 + 26.4}{5} = 25.6\,\text{cm}$

Random error = |Estimated mean – True mean| = 25.6 – 25.5 = 0.1 cm

Hence, option (D) is correct.

Q.7. In a hypothesis testing, the mean height of the population of an institute is given by 150 cm. Denote the mean as 'm', 'H_o'as null hypothesis and 'H_α'as alternate hypothesis. Which of the following option(s) is/are correct?

(A) H_o: m = 150 cm (B) H_α: m = 150 cm
(C) H_α: m < 150 cm (D) H_α: m > 150cm

Sol. The null hypothesis states that the population mean equals the specified value. The alternative hypothesis does not use equality. If the test aims to determine whether the mean is less than 150 cm, this alternative hypothesis is appropriate. If the objective is to test whether the mean is greater than 150 cm, this alternative hypothesis is correct. Hence, options (A), (C) and (D) are correct.

Q.8. Weights of male and female cadets of NCC are measured in random manner to test the null hypothesis that gender does not affect the weight. The weights of 82 female cadets and 108 male cadets were measured. The degree of freedom in hypothesis testing is

(A) 81 (B) 107 (C) 187 (D) 188

Sol.

Given,
Number of female cadets, $n_1 = 82$
Number of male cadets, $n_2 = 108$

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To calculate the degrees of freedom for hypothesis testing when comparing the means of two independent samples (e.g., female cadets and male cadets), two-sample *t*-test is used. For a **two-sample *t*-test**, the degrees of freedom can be calculated as

$$\text{Degree of freedom} = n_1 + n_2 - 2 = 82 + 108 - 2 = 188$$

Hence option (D) is correct.

Q.9. A random sample of 70 items is chosen from a lot of 2000 items. The lot contains 70 defective items. As per Poisson distribution, the probabilities of getting 0 and 1 defective items in the sample are

(A) 0.074 and 0.036, respectively
(B) 0.205 and 0.036, respectively
(C) 0.074 and 0.256, respectively
(D) 0.205 and 0.256, respectively

Sol.

Given,
Lot size, $N = 2000$
Sample size, $n = 70$

The percent defective (p) = 70/1000 = 0.07

As per Poisson distribution, the probability of acceptance is expressed as

$$P_A = \frac{(n \times p)^c \, e^{-(n \times p)}}{c!}$$

Then, probability of getting 0 defective item is

$$P_A(0) = \frac{(70 \times 0.07)^0 \, e^{-(70 \times 0.07)}}{0!} \approx 0.074$$

and, probability of getting 1 defective item is

$$P_A(1) = \frac{(70 \times 0.07)^1 \, e^{-(70 \times 0.07)}}{1!} \approx 0.036$$

Hence, option (A) is correct.

Q.10. Which of the following statement(s) is/are TRUE?

(A) AQL represents the maximum proportion of defectives which the consumer finds definitely acceptable.
(B) The *producer's risk* (denoted by α) is defined as the probability that lots of *good* quality level will *not* be accepted.
(C) The *consumer's risk* (denoted by β) is the probability that lots of non-acceptable quality level will be accepted, which otherwise should be rejected.
(D) The AQL represents the average percent defective in the outgoing products after inspection

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Sol. AQL represents the maximum proportion of defectives which the consumer finds definitely acceptable. The *producer's risk* (denoted by a) is defined as the probability that lots of *good* quality level will *not* be accepted. The *consumer's risk* (denoted by β) is the probability that lots of quality level will be accepted which otherwise should be rejected. The AOQ represents the average percent defective in the outgoing products after inspection, not AQL. Hence, options (A), (B) and (C) are correct.

Q. 11. A company uses an acceptance sampling plan to inspect a lot of products. The probability of acceptance is 0.85, the fraction defective in the lot is 0.04, the sample size is 100, and the lot size is 1000, the Average Outgoing Quality (AOQ) is

(A) 2.75% (B) 3.06% (C) 3.50% (D) 4.25%

Sol.

Given,
Probability of acceptance, $P_a = 0.85$
Defective fraction, $p = 0.04$
Sample size, $n = 100$
Lot size, $N = 1000$

Average Outgoing Quality (AOQ) = $P_a p\left(\frac{N-n}{N}\right)\times 100$

$= 0.85\times 0.04\times\left(\frac{1000-100}{1000}\right)\times 100 = 3.06$

Hence, option (B) is correct.

Q.12. In an acceptance sampling plan, a company inspects 50 items randomly from a lot. If the number of defective items is 3 or less, the lot is accepted; otherwise, it is rejected. This is an example of:

(A) Double sampling plan (B) Sequential sampling plan
(C) Single sampling plan (D) Continuous sampling plan

Sol. In a single sampling plan, only one sample is taken from the lot, and the decision to accept or reject the lot is based on that single sample. Hence, option (C) is correct.

Q.13. The first person to apply statistics in quality control is

(A) W.E. Deming (B) J.M. Juran (C) Walter A. Shewhart (D) Philip B. Crosby

Sol. Walter A. Shewhart is known as the father of Statistical Quality Control (SQC). He introduced the concept of control charts while working at Bell labs, which became a foundational tool in quality management and process control. Credit for applying statistical concepts to problems of quality control goes to Walter Shewhart.

Hence, option (C) is correct.

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Q.14. A quality engineer collected 5 samples, each of size 4, from a manufacturing process. The sample means are: 18, 20, 19, 21, 22. The average range is $\bar{R} = 4$ For sample size $n = 4$, the constant $A_2 = 0.729$. The Upper Control Limit (UCL) of the $\bar{X}$-chart is

(A) 22.92 (B) 21.92 (C) 20 (D) 17.71

Sol. The overall average can be calculated as, $\bar{\bar{X}} = \dfrac{18+20+19+21+22}{5} = 20$

Now, The Upper Control Limit, $\text{UCL} = \bar{\bar{X}} + A_2\bar{R} = 20 + (0.729 \times 4) = 22.92$

Hence, option (A) is correct.

Q.15. A manufacturing company wants to monitor the number of defective items in samples of fixed size collected every day. The most appropriate control chart is

(A) p-chart (B) c-chart (C) np-chart (D) $\bar{X}$-chart

Sol. A p-chart is used for the proportion/fraction defective. The c-chart is used for the number of defects (nonconformities) per unit. np-chart is used for the number of defective items when the sample size is constant. The $\bar{X}$-chart is used for variable data such as weight or length. Hence, option (C) is correct.

Q.16. A quality inspector takes samples of size 100 from a production process. The average fraction defective is 0.08. The Upper Control Limit (UCL) and Lower Control Limit (LCL) are

(A) UCL = 0.161, LCL = 0
(B) UCL = 0.010, LCL = 0.016
(C) UCL = 0.161, LCL = −0.001
(D) UCL = 0.180, LCL = 0.040

Sol.

Given,

samples of size, $n = 100$

fraction defective, $\bar{p} = 0.08$

For a $\bar{p}$-chart,

$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.08 + 3 \times \sqrt{\frac{0.08 \times (1-0.08)}{100}} = 0.161$$

$$\text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.08 - 3 \times \sqrt{\frac{0.08 \times (1-0.08)}{100}} = -0.001$$

As LCL is negative, it will be updated to zero.

Hence, option (A) is correct.

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Q.17. In a design of experiments, there are six parameters of influence. Each parameter takes on two levels— low and high. The number of experiments in a full factorial design is

(A) 8 (B) 12 (C) 36 (D) 64

Sol.

Given,

Number of parameters (factors) = 6

Level per factor = 2

In a full factorial design, the total number of experiments required is calculated as

$$\text{Total experiments} = (\text{number of levels})^{\text{number of factors}} = 2^6 = 64$$

Hence, option (D) is correct.

Q.18. Who is often credited as the "Father of Six Sigma"?

(A) W. Edwards Deming **(B)** Joseph M. Juran **(C)** Bill Smith **(D)** Philip B. Crosby

Sol. Bill Smith is widely regarded as the "Father of Six Sigma" for developing the methodology at Motorola in the 1980s to improve quality and reduce defects. Hence, option (C) is correct.

Q.19. With respect to Six Sigma belt hierarchy, which of the following statement (s) is/are correct?

(A) White Belt professionals possess a basic understanding of Six Sigma concepts.

(B) Yellow Belt professionals typically work as team members in Six Sigma projects.

(C) Green Belt professionals can lead projects using the DMAIC methodology.

(D) Master Black Belts are necessarily the heads of the quality control department.

Sol. White Belt holders have a basic understanding of Six Sigma concepts and terminology but are not usually involved deeply in projects. Yellow Belt professionals assist in Six Sigma projects and often work as team members supporting improvement initiatives. Green Belt professionals typically lead smaller projects and apply the DMAIC (Define, Measure, Analyze, Improve, Control) methodology. Master Black Belt professionals are highly experienced experts who guide strategy and provide mentoring. Hence, options (A), (B) and (C) are correct.

Q.20. A process produces 15000 units. Each unit has 3 defect opportunities. If 94 defects are found in total, what is the DPMO?

(A) 2089 (B) 5000 (C) 1590 (D) 18800

Sol.

Given,

Total units = 15000

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Total defects = 94
Opportunities per unit = 3

$$\text{Defects per Million Opportunity (DPMO)} = \frac{\text{Total Defects}}{\text{Total units} \times \text{Opportunities per unit}} \times 1000000$$

$$= \frac{94}{15000 \times 3} \times 1000000 = 2088.9 \approx 2089$$

Hence, option (A) is correct.

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